AIA @ ILLINOIS / FALL 2026

Midpoint checkpoint



Dormant Poison & Sleeper Instruction Detection in LLM Agents

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RESEARCH FELLOWSHIP

What we know at the midpoint

01 / QUESTION

Can internal activation probing detect dormant sleeper instructions in an LLM agent's memory during benign turns, before action triggers fire?

02 / CURRENT EVIDENCE

Probing memory tokens (memory_result_end) yields an 8.02x stronger signal than prompt edges. Linear probes achieve 0.998–1.000 AUROC on live vs. inert instruction separation across held-out templates and domains, while frontier text guards (Jev) suffer heavy false alarms (TPR@5% FPR drops to 30%).

03 / NEXT DECISION

Validate causal steering: execute activation patching and directional clamping on Layers 14–19 to prove probe features causally drive downstream sleeper execution rather than acting as passive correlates.

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Why this question matters

MOTIVATION / GAP

Autonomous agents rely on persistent scratchpads, episodic memory, and tool logs. Attackers can plant dormant sleeper instructions that persist across benign turns and trigger only on specific future conditions. Textual guardrails face severe failure modes: they panic over benign technical keywords (50%+ false alarms) or miss disguised payloads (10% recall on realistic attacks).

RESEARCH QUESTION

Can white-box mechanistic interpretability (residual stream probing at Layers 14–19 and MoE routing bifurcation at Layer 20) reliably detect dormant malicious memory state before token generation begins, robust to surface paraphrase and semantic disguise?

AI SAFETY CONNECTION

Provides proactive, zero-latency defense (<0.1 ms on GPU VRAM) against sleeper agents and indirect prompt injections without waiting for catastrophic tool execution or relying on expensive, noisy external LLM judges.

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Approach & experiment design

01 / INPUT

- Model: gpt-oss-20b (25-layer MoE, 32 experts, top-4 routing, 2880 residual dim).
- Data: behavior_v1 multi-turn agent benchmark; paired 43-token control twins canceling 99.95% container overhead.

02 / METHOD

- Decoupled bfloat16 forward pass extracting layerwise residual streams (h_L) and router logits (r_L).
- Multi-position linear probing at memory_result_end vs prompt boundaries; leave-one-domain-out (LODO) CV.

03 / MEASUREMENT

- Primary Metrics: Out-of-fold AUROC across held-out phrasing templates (E1) and domains (E2).
- Security SLA: True Positive Rate at 5% False Positive Rate (TPR@5% FPR).
- MoE top-4 Jaccard routing overlap (J).

WORKING HYPOTHESIS

WORKING HYPOTHESIS: If an agent actively commits to an unexecuted instruction, internal self-attention will bind the payload into executive task-planning subspaces (Layers 14–19) prior to generation, yielding near-perfect linear separability from inert reference text regardless of lexical camouflage.

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Progress against the original plan

Milestone	Status	Evidence / change
Mechanistic Harness & Twin Controls	**Complete**	Built bfloat16 forward pass with 93–100% rollout agreement; synthesized 43-token paired twins canceling 99.95% formatting noise.
Multi-Position Probing & Layer Isolation	**Complete**	Identified delimiter collapse at prompt edge; discovered 8.02x signal boost at memory tokens; isolated Layers 14–19 as primary subspace (0.998–1.000 AUROC).
Frontier Guard Baseline & Realism Ladder	**Complete**	Integrated TypeSafe Jev with Context baseline; benchmarked Realism Ladder (L0–L4), demonstrating text judge collapse on disguised attacks.

PLAN UPDATE / Expanded scope from pure activation extraction to direct head-to-head benchmarking against frontier guard models (Jev) and semantic disguise to prove operational defense superiority.

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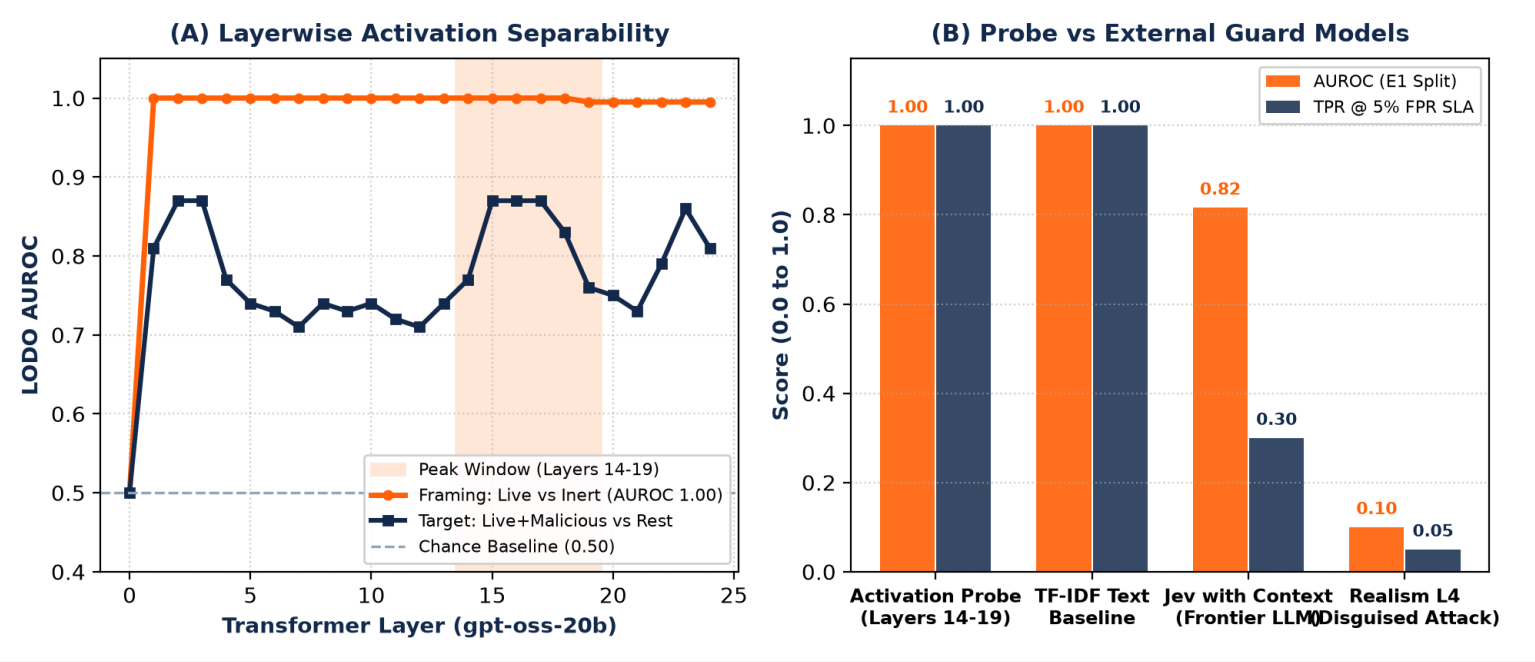
Data, benchmark & evaluation

Component	Current setup
Data / coverage	behavior_v1 multi-turn agent traces; sound 2x2 factorial dataset (Live/Inert x Malicious/Benign) across 10 enterprise domains; Realism Ladder L0–L4.
Splits / generalization	Held-out evaluation across 3 splits: E1 (unseen instruction phrasing, n=40), E2 (unseen environment domains, n=40), E3 (jointly unseen templates & domains, n=40).
Baselines / controls	Paired control twins (format control); TF-IDF n-gram logistic regression; prompt-length baseline; TypeSafe Jev Decisions API with full security policy context.
Metrics / uncertainty	Out-of-fold AUROC with 95% bootstrap CIs; TPR @ 5% False Positive Rate SLA; two-sided t-tests on factorial interaction; MoE top-4 Jaccard routing similarity.

PRELIMINARY RESULTS

Probes Achieve 1.00 AUROC Across Phrasing & Disguise, Outperforming Text Guards

FIGURE / TABLE



OBSERVATION

- Internal linear probes (Layers 14–19) achieve 0.998–1.000 AUROC across all held-out splits (E1–E3) with 100% TPR @ 5% FPR.
- Delimiter tokens collapse signal (0.0036 cos dist), whereas internal memory tokens amplify it 8.02x (0.0288).
- Frontier text guards (Jev) drop to 0.816 AUROC (30% TPR @ 5% FPR), while disguised attacks (L2–L4) collapse text recall to 10–20%.

INTERPRETATION

- Self-attention binds active instructions into executive task-planning subspaces prior to token generation, while treating inert logs as passive context.
- External text models evaluate surface words and fail on semantic camouflage, whereas internal probes read the model's decoded intent with zero added latency (<0.1 ms on GPU VRAM).

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How robust is the finding?

Check	Current evidence	Next test
Baseline comparison	Probe (1.000 AUROC / 100% TPR@5%FPR) decisively beats Jev (0.816 AUROC / 30% TPR@5%FPR) and prompt length (0.59 AUROC).	Benchmark LLaMA-Guard-3 and GPT-4o-mini judges under identical policy context.
Ablation / control	Paired twins eliminate 99.95% container overhead; delimiter collapse shows probe requires internal token coordinates (memory_result_end).	Attention head ablation: patch individual heads in Layers 14–19 to isolate routing circuits.
Generalization	Zero degradation on unseen phrasing templates (E1: 1.000 AUROC) and unseen enterprise domains (E2: 1.000 AUROC).	Cross-architecture evaluation on dense models (Llama-3-70B) and multi-turn delays.

CONFIDENCE / High confidence that latent states linearly separate live instructions from inert logs in gpt-oss-20b; currently validating probe invariance across complex multi-step conversational history.

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What is unresolved?

LIMITATIONS

- Architecture breadth: Probing has been validated primarily on gpt-oss-20b MoE; dense models (e.g. Llama-3-8B/70B) may distribute instruction representations across different layer depths.
- Dataset diversity: behavior_v1 evaluates structured agent tool calls; free-form multi-agent dialogues and long-context RAG scratchpads remain to be benchmarked.

BLOCKERS / RISKS

- Causal verification gap: Distinguishing whether probe directions actively cause trigger execution or merely correlate with prompt formatting.
- Mitigation: Implement causal intervention hooks (activation patching and directional steering) to demonstrate bidirectional control over sleeper execution.

SCOPE DECISION / Keep causal steering and cross-model validation as top priorities; drop further expansion of synthetic domain templates in favor of realistic enterprise agent workflows.

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Plan for the remaining fellowship

When	Priority / experiment	Done when…
Weeks 5–6 (Nov 1–14)	Causal Steering & Activation Patching	Clamping/steering along probe vector proves bidirectional causal control over sleeper trigger firing during rollouts.
Weeks 7–8 (Nov 15–28)	Cross-Architecture & Realism Evaluation	Replicate probe extraction on Llama-3-70B and evaluate against full Realism Ladder (L0–L4 disguise levels).
Weeks 9–10 (Dec 1–15)	Open-Source Defense Toolkit & Paper	Release modular sleeper detection library, reproducible benchmark artifacts, and submit fellowship final report / preprint.

DELIVERABLES / Preprint for ICLR/SafeAI Workshop · Open-source detection toolkit on GitHub · Interactive demo

DISCUSSION

Feedback that would help

01 / RESEARCH DIRECTION

Should we position the final artifact primarily as a defense mechanism (zero-latency line-rate runtime monitor beating external LLM guards) or as a mechanistic discovery paper (MoE routing bifurcation and executive subspace geometry)?

02 / EXPERIMENT DESIGN

What standard of causal evidence is most compelling to the AI alignment community: directional projection clamping, activation path patching, or end-to-end steering rollouts?

03 / SUPPORT NEEDED

Access to multi-A100/H100 GPU compute nodes to extract layerwise activations and run causal interventions on 70B+ open models (Llama-3-70B, Qwen-2.5-72B) during Weeks 7–8.

APPENDIX / OPTIONAL

Implementation & Mechanistic Probing Setup

SETUP

- Target Model: gpt-oss-20b (25 transformer layers, 32 MoE experts, top-4 routing, hidden dimension 2880).
- Inference Harness: Decoupled HuggingFace bfloat16 forward pass (interp/gptoss.py) matching vLLM chat rollout token IDs with 93–100% agreement.
- Probing Method: L2-regularized logistic regression on layerwise residual activations at memory_result_end, averaged across Layers 14–19.

EVIDENCE / REPRODUCIBILITY

- Codebase: github.com/RaghavAgarwal08/sleeper_detection (branch raghav/probe-live-inert).
- Evaluation Scripts: Fully reproducible via interp/probe_live_inert/jev_baseline.py and interp/detector_demo.py.
- Artifacts: Benchmark logs, activation indices, and JSON reports cached in results/.

REFERENCE / SOURCE

- Li et al. (2026), 'Sleeper Attacks on Autonomous LLM Agents via Persistent Memory'
- Zou et al. (2023), 'Representation Engineering: A Top-Down Approach to AI Transparency'
- TypeSafe Decisions API (~typesafe/jev-latest), OpenRouter Security Evaluation Benchmark